Sebastian Tamayo

Senior Software Engineer

+57 3113698653• sebstiantamayo629@outlook.com• https://www.linkedin.com/in/sebastian-tamayo-pro/ • Pereira, Colombia

Results-driven Senior Web Developer with over a decade of experience designing, developing, and deploying scalable, high-performance web applications with a focus on AI-driven solutions and enterprise systems. Specializes in building modern, responsive frontends using React.js and Next.js, and architecting robust backends with Node.js, Express.js, and NestJS (TypeScript), integrating SQL/NoSQL databases such as MySQL and PostgreSQL. Proficient across the entire software development lifecycle—from requirements gathering and system architecture to CI/CD deployment, automated testing, monitoring, and long-term maintenance. Experienced in designing AI/ML-powered features, including natural language processing, predictive insights, document summarization, and intelligent chatbots, leveraging Azure OpenAI Service, Azure Cognitive Services, and Hugging Face models. Thrives in Agile, cross-functional teams, collaborating with product owners, designers, legal SMEs, and engineers. Committed to Responsible AI practices, ensuring fairness, transparency, safety, and privacy in AI solutions, while maintaining enterprise-grade security and compliance standards. Known for writing clean, maintainable, and well-documented code, mentoring junior developers, and promoting SOLID principles and clean architecture standards. Passionate about performance optimization, system scalability, and creating high-impact digital products that deliver measurable value to users and business stakeholders. Continuously learning and staying current with emerging AI technologies, cloud infrastructure (Azure & AWS), and DevOps practices.

PROFESSIONAL EXPERIENCE

UST

Senior Software Engineer

June 2022 – June 2025 (3 years)

- Collaborated within a cross-functional Agile Scrum team, mentoring junior developers and conducting peer code reviews, applying SOLID principles, clean architecture, and design patterns to ensure maintainability, scalability, and reliability.
- Architected and re-engineered a customer-facing web platform using React.js, Next.js, Redux, and TypeScript, integrating APIs, telemetry, and real-time features that enhanced user satisfaction, increased engagement, and improved conversion rates.
- Designed and implemented scalable backend services with Node.js, Express.js, NestJS, and C#/.NET, persisting data in MySQL/SQL Server and optimizing queries with TypeORM, achieving measurable performance improvements.
- Built and deployed microservices architectures using Docker and AWS ECS/Azure Kubernetes Service (AKS), enabling modular independence, high availability, and asynchronous communication through RabbitMQ and Azure Service Bus.
- Developed an AI-powered chatbot and legal document assistant leveraging Azure OpenAI, Hugging Face transformers, and NestJS, improving automated support, natural language search, and predictive insights in sensitive legal contexts.
- Conducted AI/ML experiments and prototypes (LLM fine-tuning, embeddings, semantic search, summarization), validating ideas and integrating them into production systems with Azure Cognitive Services and Azure Machine Learning.
- Refactored frontend components and backend logic to reduce latency by 30% and enhance system resiliency through improved error handling, observability, and monitoring dashboards (Grafana, Azure Monitor).
- Implemented and maintained CI/CD pipelines with GitHub Actions, integrating infrastructure-as-code (Terraform/Bicep), automated testing (Jest, Cypress), and security checks to shorten release cycles and ensure enterprise-grade reliability.

- Actively participated in security and privacy reviews, performing threat modeling, Responsible AI assessments, and privacy impact analyses (PIAs) to ensure solutions adhered to fairness, transparency, and compliance standards.
- Designed engineering KPIs and monitoring systems to track application reliability, drift in AI models, and user feedback, enabling continuous improvement and Responsible AI governance.

Tech Stack:

React.js, Redux, Next.js, Node.js, Express.js, Azure OpenAI, Hugging Face transformers, GraphQL, REST API, MongoDB, MySQL, PostgreSQL, Redis, TypeORM, Prisma, NestJS, Shopify, AWS (Lambda, ECS, API Gateway, AppSync, S3, IVS, DynamoDB, CloudWatch, CloudFormation, IAM), Docker, Terraform, Vercel, Cypress, Jest, Playwright, MSW, RabbitMQ, WebSockets, Ably, Git, GitHub Actions, Postman, Figma, Notion, Jira, Adobe XD, Google Analytics, LogRocket, Sentry, AI/ML/NLP (OpenAI, AWS Comprehend, AWS Lex, Hugging Face, spaCy)

Gorilla Logic

Full Stack Engineer

February 2018 – May 2022 (4 years 4 months)

- Engineered responsive, mobile-optimized UIs with React.js, Next.js, TypeScript, and Tailwind CSS, achieving a 40% increase in mobile user retention and a 30% boost in engagement, while ensuring accessibility and compliance with enterprise design standards.
- Developed modular, component-driven dashboards and legal-tech tools, leveraging Redux and clean architecture principles to deliver scalable, maintainable, and reusable components across applications.
- Built and maintained backend services with Node.js, NestJS (TypeScript), and C#/.NET, implementing microservices and event-driven architectures with Kafka/Azure Service Bus, enabling reliable, asynchronous workflows.
- Provisioned cloud infrastructure with AWS CloudFormation and Azure Bicep/Terraform, deploying containerized services via Docker to AWS ECS and Azure Kubernetes Service (AKS), reducing deployment cycle time by 50% and improving scalability.
- Delivered real-time analytics and AI-driven insights pipelines by orchestrating AWS Lambda, Azure Functions, Cognitive Services, and API integrations, enabling predictive modeling, natural language search, and contract summarization.
- Optimized SQL/NoSQL databases (MySQL, PostgreSQL, DynamoDB) by refactoring schemas and queries with TypeORM, improving database response times by 15% and ensuring scalability for AI-driven workloads.
- Orchestrated services with Kubernetes (EKS/AKS) to ensure high availability, self-healing, and seamless horizontal scaling of mission-critical applications in production.
- Implemented Responsible AI and security best practices, conducting threat modeling, privacy impact assessments (PIAs), and monitoring model drift to safeguard sensitive legal and compliance-related data.
- Followed TDD practices with Jest and Jasmine, integrated CI/CD pipelines with GitHub Actions, and contributed to observability strategies using AWS CloudWatch, Azure Monitor, and APM tools, strengthening reliability and transparency.

Tech Stack:

React.js, Redux, Next.js, TypeScript, Tailwind CSS, Node.js, NestJS, C#/.NET, Kafka, REST API, GraphQL, TypeORM, MySQL, PostgreSQL, DynamoDB, Docker, Kubernetes (EKS/AKS), AWS (Lambda, S3, ECS, CloudFormation, CloudWatch), Azure (OpenAI, Cognitive Services, Functions, Machine Learning, Bicep), Ably, WebSockets, Jest, Jasmine

Sophos Solutions

5 years 6 months

Senior Web Developer

September 2015 – February 2018 (2 years 6 months)

- Contributed to the development of a high-performance Single Page Application (SPA) for the financial sector using React.js, Next.js, and Redux, achieving a 30% improvement in user engagement and application responsiveness.
- Refactored and modularized legacy codebases while building scalable RESTful backend APIs with Node.js, and Express.jsresulting in a 40% reduction in server response time and improved code maintainability.
- Applied advanced frontend optimization techniques—including code splitting, tree shaking, minification, image compression, and CSS sprite maps—reducing initial load time by 35% and improving SEO rankings to the top 5 in Google search results.
- Developed mobile-first and AMP-compliant user experiences, increasing mobile traffic by 25%. Leveraged localStorage and sessionStorage to enhance client-side performance and persistence.
- Implemented automated testing pipelines using Jest (unit) and Cypress (E2E), reducing post-deployment bugs by 75% and streamlining deployments via Git-based CI/CD workflows.
- Collaborated closely with UX/UI teams to redesign complex enterprise dashboards in Next.js and Redux, improving navigation speed by 50% and boosting internal productivity by 25%.

Tech Stack:

React, Next, Redux, Angular, RxJS, TypeScript, JavaScript, Node.js, Express.js, PHP, Laravel, Shopify, WordPress, REST API, MySQL, Jest, Cypress, Git, CI/CD, AMP, Adobe XD

Web Developer

September 2012 – February 2015 (2 years 6 months)

- Built responsive, standards-compliant websites using HTML5, CSS3, JavaScript (ES6+), and SASS, aligned with modern Web 2.0 usability and accessibility standards.
- Designed and implemented reusable, testable components using React and Backbone.js, applying best practices in modular architecture and dependency injection.
- Built scalable REST-based web services with Node.js and Express.js, applying asynchronous programming patterns and middleware design for high-performance network applications.
- Employed Gulp, Grunt, and NPM scripts to automate build pipelines, handle asset bundling, preprocess stylesheets, and manage static resources for faster development cycles.
- Collaborated closely with backend teams to integrate APIs, caching layers, and proxy configurations, ensuring seamless CI/CD deployment pipelines with tailored build/test scripts.
- Developed advanced PL/SQL stored procedures and complex queries using temporary tables and procedural logic to support dynamic reporting and data transformation requirements.
- Integrated Google Analytics and custom tracking solutions to gather behavioral insights, which informed UX decisions and resulted in improved conversion funnels.
- Reduced page load time by 35% using frontend performance optimization techniques—such as JS/CSS minification, lazy loading, image compression, and CSS sprites—boosting SEO rankings to achieve a top 5 position on Google SERPs.

Tech Stack:

React, HTML5, CSS3, JavaScript (ES6+), ECMAScript, SASS, AngularJS (1.x, 2.x, 4.x), Backbone.js, Node.js, Express.js, REST API, Gulp, Grunt, NPM, SQL, PL/SQL, Google Analytics, CI/CD, SEO Optimization, Git

EDUCATION

Universidad Católica de Pereira

04.2005 – 09.2012

Bachelor of Computer Science

SKILLS

Languages:	C++, JS(ES6), TS, Python, C#, Rust, PHP, Java, HTML5
Frameworks and Libraries:	React.js, Next.js, Gatesby, Angular, Node.js, Express, React Native, Redux, RxJS, REST API, GraphQL, Ably, Laravel, Tailwind CSS, RabbitMQ, Kafka, TypeORM, Shopify, WordPress, JPA, Java Spring Boot
DBMS	MySQL, MS SQL, PostgreSQL, MongoDB, AWS DynamoDB, AWS RDS, Azure SQL Database
IDEs and Tools:	Visual Studio Code, Jenkins, Cypress, Jest, Jasmine, Terraform, Docker, Kubernetes, Redux DevTools, Babel, Webpack, Storybook, Azure DevOps, Git Bash, PowerShell
Cloud Services:	AWS Cloud Services, Microsoft Azure, GCP
Other skills:	Problem Solving, Full Stack Development, Critical Thinking, Leadership, Algorithm Design, Team-Oriented, SOLID, Agile (Scrum), Microservices, Test-Driven Development (TDD), Diagnosing, Business Analysts